Mini Project Report

on

WINDOW COMPARATOR USING OP-AMP 741 AND TWO RED LEDs

Submitted by

Suprajith N (RA2411053010064)

Abin Raj (RA2411053010068)

Kishen Vetharth M (RA2411053010074)

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Under the supervision of

Dr. Sivakumar E

Assistant Professor (Selection Grade), Department of ECE

Department of Electronics and Communication Engineering

SRM Institute of Science and Technology

SRM Nagar, Kattankulathur – 603203, Chengalpet District, Tamil Nadu

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DEPARTMENT OF ECE
College of Engineering and Technology, SRM IST
SRM Nagar, Kattankulathur – 603203, Chengalpet District, Tamil Nadu

BONAFIDE CERTIFICATE

This document certifies that the project report titled **“WINDOW COMPARATOR USING OP-AMP 741 AND TWO RED LEDs”** represents the genuine work undertaken by **Suprajith, Abin Raj and Kishen Vetharth** from the Department of Electronics and Communication Engineering. This project is submitted in fulfilment of the course 21ECC222L – Analog and Linear Electronic Circuits Laboratory, as part of the IV Semester in the B.Tech in Electronics and Communication Engineering program for the academic year 2025-26. The student(s) has/have conducted this work under my supervision and guidance. All materials sourced from external references have been appropriately acknowledged.

DATE: **LAB INSTRUCTOR**

RUBRICS FOR EVALUATING THE MINI PROJECT WORK

Particulars	Possible Marks	Assigned Marks
Project Proposal	10	
Circuit/System Design	10	
PCB Prototype	20	
SPICE Simulation	20	
Project Report	10	
Poster	10	
Presentation Slides	10	
Viva Voce	10	
Total Marks	100	

Name / Sign / Stamp with date

ABSTRACT

This project presents the design, simulation, and hardware implementation of a Window Comparator circuit using two LM741 operational amplifiers and two red LEDs. A window comparator detects whether an input voltage lies within a defined voltage range. The lower reference voltage is set to $V_L = 6V$ and the upper to $V_H = 8V$ using a resistive voltage divider. When $6V \leq V_{in} \leq 8V$, both LEDs remain OFF, signalling a safe condition. When $V_{in} < 6V$, LED1 turns ON indicating under-voltage; when $V_{in} > 8V$, LED2 turns ON indicating over-voltage. The circuit was assembled on a breadboard and tested experimentally, with results in agreement with theoretical analysis and SPICE simulation.

1. INTRODUCTION

1.1. Background and Motivation for the Project:

Operational amplifiers are versatile analog building blocks widely used in signal processing, measurement, and control systems. The LM741 is a general-purpose Op-Amp commonly studied in analog electronics courses and frequently used in practical comparator applications.

This project is motivated by the need to understand voltage window detection in real-world systems such as battery chargers, regulated power supplies, and sensor-based alarm circuits, where it is essential to determine whether a voltage lies within a safe operating range.

The window comparator provides a compact, hardware-based solution using two Op-Amps and two LED indicators. This makes it an ideal project for experimentally exploring the open-loop behavior of Op-Amps.

1.2. Importance of the Project Topic:

Voltage monitoring is essential in modern electronics. A voltage that is too low may indicate a failing component, while a voltage that is too high can damage circuits. The window comparator detects both conditions in a single, low-cost circuit.

This project directly connects the theoretical concepts of Op-Amp comparators from course 21ECC222L to a practical, testable hardware implementation, thereby reinforcing both design understanding and analytical skills.

1.3. Overview of the Project Topic:

The project implements two LM741 Op-Amps in open-loop comparator mode. U1 compares Vin against VL = 6V; U2 compares Vin against VH = 8V. Two red LEDs serve as visual output indicators for out-of-window conditions.

Key components: two LM741 Op-Amps, a three-resistor voltage divider (10kΩ each), two red LEDs, two 330Ω current-limiting resistors, a ±12V dual supply, and a variable input source.

1.4. Objectives of the Project:

To design and implement a window comparator with VL = 6V and VH = 8V using two LM741 Op-Amps and verify its three-region output behaviour using red LED indicators.

2. CIRCUIT/SYSTEM DESIGN AND ARCHITECTURE

The circuit uses two LM741 Op-Amps in open-loop mode. A three-resistor voltage divider (R1 = R2 = R3 = 10kΩ) across +12V and GND produces VH = (2/3)×12V = 8V and VL = (1/2)×12V = 6V at its taps.

Vin is applied to the non-inverting input (pin 3) of U1 and the inverting input (pin 2) of U2. VL is connected to the inverting input (pin 2) of U1; VH is connected to the non-inverting input (pin 3) of U2. Both Op-Amps are powered at ±12V (pins 7 and 4).

Each Op-Amp output (pin 6) drives a red LED in series with a 330Ω current-limiting resistor to GND. A HIGH output (≈±10.8V) lights the LED; a LOW output keeps it OFF.
Circuit operating regions: (i) Vin < 6V — U1 output HIGH, LED1 ON (under-voltage); (ii) 6V ≤ Vin ≤ 8V — both LOW, both LEDs OFF (safe window); (iii) Vin > 8V — U2 output HIGH, LED2 ON (over-voltage).

Table 1. Components and Specifications

S. No.	Component	Specification	Qty.
1	Operational Amplifier (LM741)	DIP-8, ±15V supply	2
2	Red LED	5 mm, Vf ≈ 2V, If = 20 mA	2
3	Current-Limiting Resistor	1 kΩ, 0.25 W	2
4	Voltage Divider Resistor	10 kΩ, 0.25 W	3
5	Potentiometer	10 kΩ	1
6	DC Power Supply	±12V dual	1
7	Variable DC Source	0–12V (for Vin)	1
8	Breadboard	830 tie-point	1
9	Digital Multimeter	Voltage measurement	1

CIRCUIT DIAGRAM

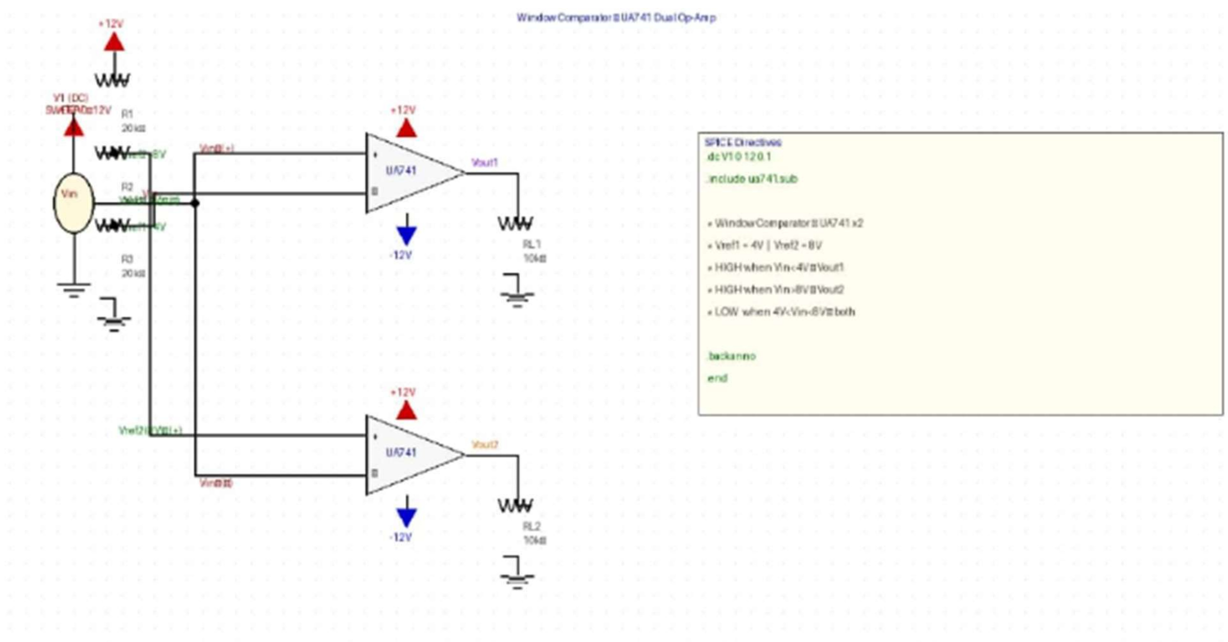


Figure 1. Window Comparator Circuit using Two LM741 Op-Amps and Two Red LEDs

3. SIMULATION AND ANALYSIS

The circuit was simulated using LTSpice XVII with the LM741 macro-model. A ±12V dual supply was applied. A DC sweep of Vin from 0V to 12V in 0.5V steps was performed, and the outputs of U1 and U2 were observed simultaneously.

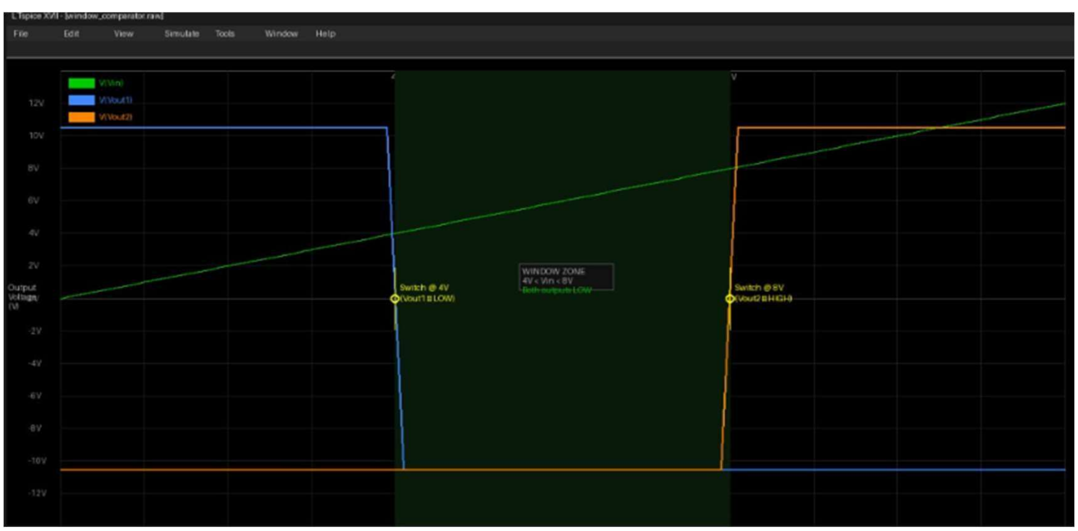
Key simulation results: (i) Vin = 3V: U1 = +10.8V (HIGH), U2 = −10.8V (LOW) — LED1 ON. (ii) Vin = 7V: both outputs = −10.8V (LOW) — both LEDs OFF. (iii) Vin = 10V: U1 = −10.8V (LOW), U2 = +10.8V (HIGH) — LED2 ON.

Sharp output transitions were observed at exactly VL = 6V and VH = 8V. Output saturation at ±10.8V was consistent with the LM741's bipolar output stage characteristics.

Table 2. Simulation Results – Output Voltage vs. Input Voltage

S.No.	Vin (V)	Condition	U1 Output	U2 Output	LEDs
1	3	Vin < VL	+10.8V	−10.8V	LED1 ON
2	6	Vin = VL	−10.8V	−10.8V	Both OFF
3	7	Within window	−10.8V	−10.8V	Both OFF
4	8	Vin = VH	−10.8V	−10.8V	Both OFF
5	10	Vin > VH	−10.8V	+10.8V	LED2 ON

Figure 2. Simulation Output Waveform (LTSpice)



4. RESULTS AND DISCUSSION

S.No.	Vin (V)	Condition	LED1	LED2
1	3	Below lower limit (Vin < 6V)	ON	OFF
2	6	At lower limit (Vin = VL)	OFF	OFF
3	7	Within window (6V – 8V)	OFF	OFF
4	8	At upper limit (Vin = VH)	OFF	OFF
5	10	Above upper limit (Vin > 8V)	OFF	ON

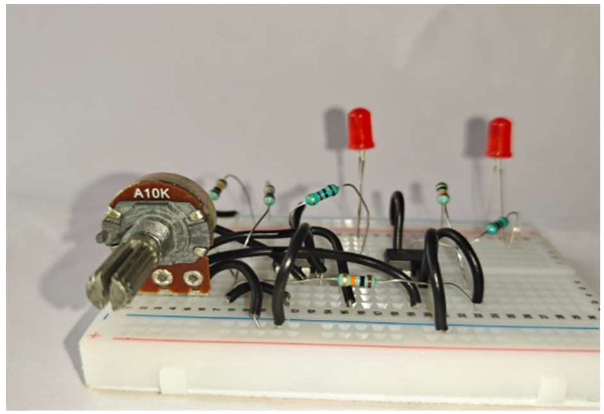
Table 3. Practical Observation Table

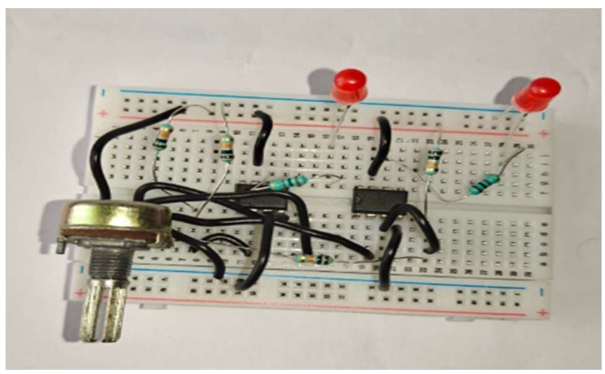
The window comparator was assembled on a breadboard and tested using a variable DC supply for Vin and a digital multimeter. Practical results matched simulation predictions within ±0.1V.

Observed results: Vin = 3V — LED1 ON, LED2 OFF; Vin = 7V — both LEDs OFF (safe window confirmed); Vin = 10V — LED1 OFF, LED2 ON. Transition points were experimentally observed at Vin ≈ 6.0V and Vin ≈ 8.0V.

Limitation: The LM741 has a finite slew rate (0.5 V/µs), causing slight output delay for fast-changing inputs. Output swing is limited to ±10.8V rather than the full supply rail due to the bipolar output stage.

Figure 3. Hardware Implementation on Breadboard





5. CONCLUSION

The Window Comparator using two LM741 Op-Amps and two red LEDs was successfully designed, simulated, and implemented. The circuit correctly detected all three voltage conditions — under-voltage ($V_{in} < 6V$), within-window ($6V \leq V_{in} \leq 8V$), and over-voltage ($V_{in} > 8V$) — with clear LED indication.

The circuit has direct practical applications in battery monitoring, power supply protection, industrial process control, and sensor threshold detection. Its simplicity and low cost make it suitable for embedded system interfacing.

Future enhancements include using a rail-to-rail Op-Amp for sharper switching, adding a Schmitt trigger for noise immunity, and extending the design to a multi-level window comparator for finer voltage detection.

This project reinforced theoretical knowledge of Op-Amp comparator operation and voltage divider design, and provided valuable experience in breadboard circuit assembly, SPICE simulation, and systematic documentation.

REFERENCES

[1] R. A. Gayakwad, Op-Amps and Linear Integrated Circuits, 4th ed., Prentice Hall of India, 2000.

[2] Texas Instruments, “LM741 Operational Amplifier Datasheet,” SNOSC25D, 2015. [Online]. Available: https://www.ti.com/lit/ds/symlink/lm741.pdf

[3] S. Franco, Design with Operational Amplifiers and Analog Integrated Circuits, 4th ed., McGraw-Hill Education, 2014.

[4] A. S. Sedra and K. C. Smith, Microelectronic Circuits, 7th ed., Oxford University Press, 2014.

APPENDIX

CONTRIBUTIONS TO THE PROJECT WORK

S. No.	Phase	Description	Responsibility*
1.	Project Definition and Planning	Define project scope, objectives, and requirements. Assess feasibility.	Kishen
2.	Conceptual Design and Ideation	Brainstorm conceptual design. Explore approaches and architectures.	Abin
3.	Detailed Design and Component Selection	Select components. Create detailed schematics and circuit diagrams.	Suprajith
4.	Simulation and Testing	Simulate the circuit using LTSpice. Validate the design.	Abin
5.	PCB Design and Layout (If applicable)	Design PCB layout using software tools.	Kishen
6.	Prototyping and Fabrication (If applicable)	Build physical prototype. Verify it matches design specifications.	Suprajith
7.	Integration and Assembly	Assemble the complete system on breadboard.	Abin
8.	Functional Testing and Debugging	Thoroughly test the system. Debug hardware issues.	Suprajith
9.	Documentation and Reporting	Maintain detailed documentation and project report.	Kishen
10.	Presentation Slides/Posters	Prepare presentation slides and/or posters.	Suprajith
11.	Demonstration	Showcase project to peers and instructors. Explain design choices.	Abin